Cyber Security and Digital Hygiene for Safe Online Ecosystems

By Shaweta Sharma (Scientist 'D')

Content of the Presentation

1) Cyber Security Fundamentals: Background, stats, challenges
2) Reals of the Cyber word, Recognizing and Defining Computer Crimes
3) Cyber Crimes, Types of Cyber Crimes, Attack Vectors
4) Knowledge of latest Tools and Techniques used in Cyber Crimes
5) D-DOS attack, Service Disruption, APTs, Common Attack techniques
6) Cyber Threat Intelligence and OSINT

The world that we live in !

Digital Era

I.O.T.

Certain incidents !

- Sony – Systems thrown into disarray; FBI investigation blamed North Korea; promptly denied
- JP Morgan Chase – Credit card data hack – 80 million households & 7 million small to medium size businesses impacted
- Chinese hacked into US weather Systems, Defence Research Systems
- iCloud – Personal and intimate pictures of celebrities went online
- US Postal service – Information pertaining to 800,000 employees compromised
- Google accounts – Thousands of users IDs and password distributed over internet
- Target Payment System – 110 million records pilfered
- European Central Bank – Employee data compromised
- Facebook, Aadhar, EVM, American Elections ???

There are two types of companies: those who have been hacked, and those who don't yet know they have been hacked.
John Chambers
Chief Executive Officer of Cisco

Observations

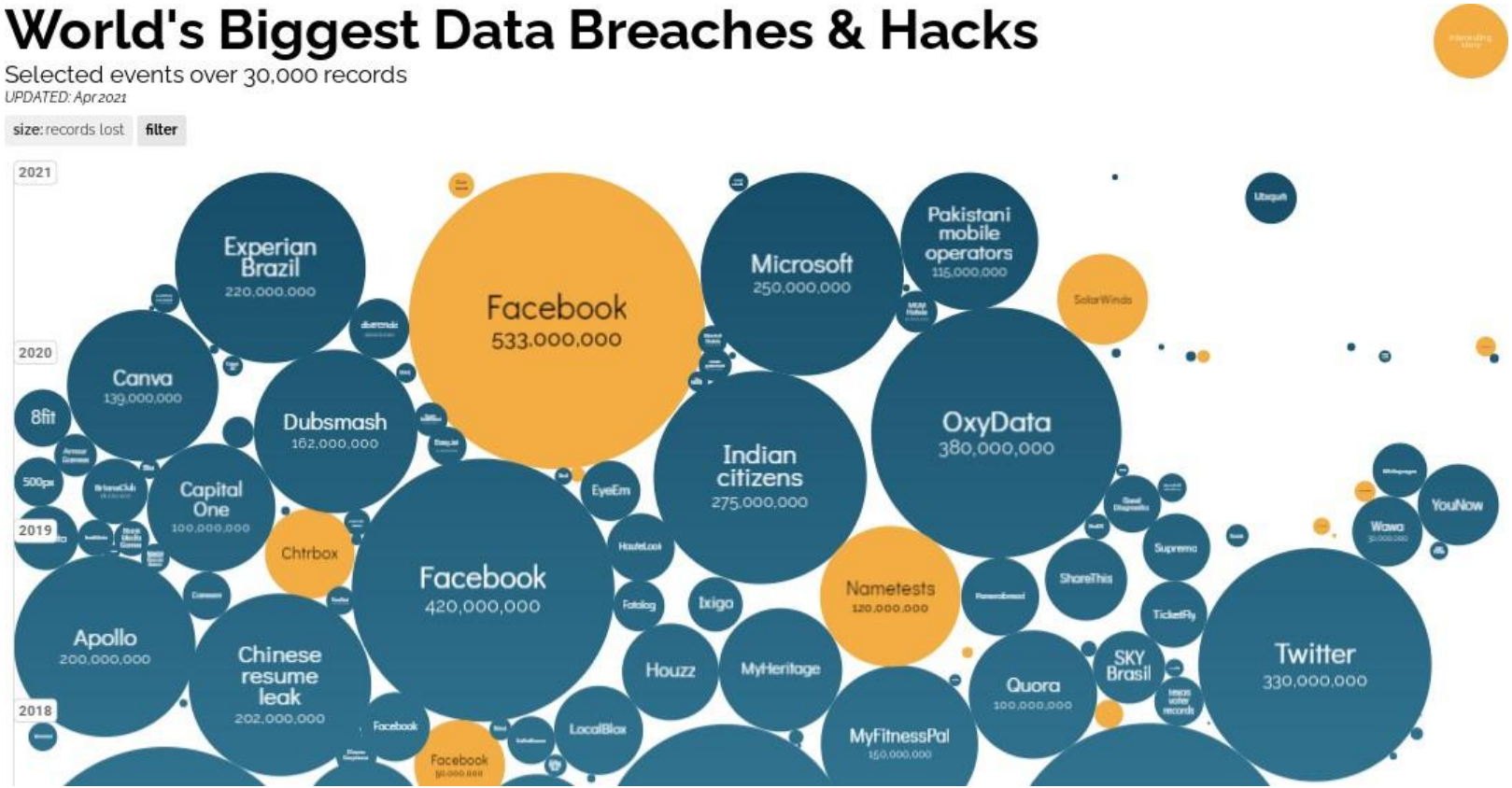
- More than half of all consumers have experienced a cybercrime, with around one in three falling victim in the past year alone. (Norton, 2021)
- Global cybercrime costs are expected to grow by 15% per year over the next five years, reaching $10.5 trillion annually by 2025. (Cybersecurity Ventures, 2020)
- The first half of 2021 saw a 102% increase in ransomware attacks compared to the beginning of 2020. (Check Point, 2021)
- The first reported death by ransomware occurred in September 2020, when a ransomware attack caused IT failure at a hospital in Dusseldorf, Germany. (Associated Press, 2020)
- Over 75% of targeted cyberattacks start with an email. (RoundRobin, 2020)
- Cyber attacks are more likely to bring down F-35 jets than missiles. (Interesting Engineering, 2021)
- A hacker accessed but unsuccessfully attempted to poison the water supply in Oldsmar, Florida, in February 2021. (Wired, 2021)
- https://www.bbc.com/news/world-us-canada-55989843

Cyber Risk Scenario

- The cyber-threats across the globe are continuously increasing. As per the Global Risks Report 2021 of the World Economic Forum, the failure of cybersecurity measures is one among the top 10 risks by likelihood.
- As per McAfee's new global report titled "The Hidden Costs of Cybercrime", the global losses from cybercrime now total over $1 trillion, a more than 50 per cent increase from 2018. In the same report, McAfee mentioned that two-thirds of surveyed companies reported some kind of cyber incident in 2019, the average interruption to operations was at 18 hours; the average cost was more than half a million dollars per incident.

The Big Picture !

http://www.informationisbeautiful.net/visualizations/worlds-biggest-data-breaches-hacks/



Live Cyber Attack Threat Map
https://threatmap.checkpoint.com/ThreatPortal/livemap.html
Powered by ThreatCloud Intelligence
THREATCLOUD
LIVE CYBER ATTACK THREAT MAP
Check Point
SOFTWARE TECHNOLOGIES, INC.
ATTACKS TODAY
(since 12AM PST)
11,016,028
ATTACKS YESTERDAY
34,619,076
TOP TARGETS BY COUNTRY
CA,USA
NJ,USA
FL,USA
France
India
Sri Lanka
Indonesia
TIME ATTACK SOURCE TARGET
21:00:51 Trojan.WIN32.XMRig.T CA,USA Indonesia
21:00:50 Trojan.WIN32.XMRig.A Netherlands India
21:00:50 Trojan.WIN32.XMRig.A France India
21:00:50 Trojan.WIN32.XMRig.T CA,USA Indonesia
21:00:50 Trojan.WIN32.XMRig.A Netherlands India
Source
Target

Cyberspace

It is an ecosystem consisting of interconnected network of digital technology ***including the Internet, telecommunications networks, computer systems, and embedded processors and controllers.***

Cyber Attack

- **Cyber Attack** is an attempt to destroy, infect or alter digital networks in order to extract or extort money or for other malicious intentions such as procuring necessary information.

- Cyber Attacks can range from installing spyware on a personal computer to attempting to destroy the infrastructure of entire nations.

Internet Uses Stats

- According to the latest report by the **Internet & Mobile Association of India (IAMAI) and Nielsen –** With over 503 million internet users, India is the second largest online market in the world, ranked only behind China. It was estimated that by 2023, there would be over 650 million internet users the country.
- ITU estimates that approximately 6 billion people – or 74 per cent of the world's population – are using the Internet in 2025. This represents an increase from 60 per cent in 2020, with 1.3 billion people estimated to have come online during that period
- It had shown rural India had 227 million active internet users, 10% more than urban India's about 205 million, as of November 2019.
- 71 Million kids between 5-11 age also go online using adult device.
- Overall gender distribution -: Men : 65% & Women : 35%
- 375 Internet and Telecom Service Providers offering the Internet, Mobile and Wireless connectivity to the large user base.

Cyber Attacks Stats

- According to **MeITY** about nearly **7 lakh** cyber attacks happened in 2020 and India is among the top five countries with the greatest number of cyber-attacks.

- India secures a spot amongst the top 10 spam-sending countries in the world alongside USA.

- As per information tracked by the Indian Computer Emergency Response Team (CERT-In), 172 and 105 government websites, which includes 74 and 31 websites hosted on NICNET managed by National Informatics Centre (NIC) were hacked during the year 2017 and 2018.

- India sees 37% increase in data breaches in 2020. The recent data breach at BigBasket, in which the details of around **2 crore** users were put on sale on a forum in the dark web.

Some Minor to Major Cyber attack in India

SIM Swap Fraud

In August 2018, two men were involved in fraudulent activities concerning money transfers of **4 Crores** from the bank accounts of numerous individuals by getting their SIM card information through illegal means.

Cyber Attack on Cosmos Bank

In August 2018, cyber attack on Cosmos Bank's Pune Branch which saw nearly **94 Crores** rupees being siphoned off Hackers hacked into the ATM server of the bank and stole details of many visa and rupee debit card owners.

Phishing Attack on Wipro

In December 2018, Phishing attack was done by a hacker group through gift card fraud. Many employees and client accounts were compromised.

Juspay Data for sale on dark web

In January2020, Details of close to **35 million** customer accounts, including masked card data and card fingerprints, were taken from a server and put for sale on dark web.

Malware attack on Nuclear power plant

In October 20 2019, authorities confirmed that **India's largest nuclear power station in Kudankulam** fell victim to a cyber spy attack, by the North Korean hacker group. They were allegedly willing to grab information on thorium-based reactors.

Who are these people

- Hackers : Black, white , grey !
- Curious Kids or enthusiasts
- Professional Criminals – Group of experienced hackers
- Malware programmers
- Corporate Espionage
- Information warfare
- Cyber Terrorism
- State Players
- Stake holders – Employees, Vendors?

It’s a rising trend

- 250,000 new malware pieces released daily
- 30,000 new websites infected daily
- More than 80% of them are Small Business sites
- 76% Indian businesses hit by cyber attacks in 2021
- India third most prone country to be hit by Cyber attacks

Reasons for this trend

Expanding Surface Area:

- Tremendous growth in usage of IT - Globalization of IT and information systems with high-speed internet connections
- Expanding wireless connectivity and multiplicity of wireless devices.
- Widespread use of e-mails.
- Increased web access to a wide range of web services and web applications & Cloud platform
- Tremendous Growth of Social Media – Facebook, WhatsApp, Twitter, YouTube, etc.

Cyber threat environment :

- Threat vectors (Attackers & Tools) - High sophistication, very focussed and rapidly evolving.
- Ease of Use of attack technologies & Proliferation of attack tools.

Cyber Security

- Cyber Security is the body of technologies, processes and practices designed to protect digital technology like networks, computers, programs and data from attack, damage or unauthorized access.

Evolution of Cyber Security

1980s
Virus

AntiVirus

1990s
Worms, Ransomeware

Intrusion detection System

2000s
Web Vulnerabilities

Web Security

2010s
Data breach,DDOS

Data Loss Protection, Application-ware firewall, Security information management

Current
Cyber warefare

IDS/IPS using ML and AI

Challenges to Cyber Security

- Fast and constantly evolving nature of security risks.
- Ever evolving standards, technologies, services, applications like AI ,ML ,quantum computing , Block Chain etc.
- Increasing complexity of systems.
- Lack of uniformity in devices used for internet access .
- Sensors, Appliances and Vehicles are also joining the Internet using IOT.
- According to Cyber security Status Report released by **ISACAA** , there is **shortage** of cyber security professionals in India.
- Lack and late national wide awareness among citizens about securing their personal information.

Basic of Information Security

CIA Triangle form the cornerstone of any organization's security infrastructure; in fact, they should function as goals and objectives for every security program.

Basic of Information Security: CIA

Information is meaningful data which has to be protected in order to protect the privacy, security, identity of an organization or a person or a nation. An information is called valuable because of few characteristics.

Confidentiality

Confidentiality is about preventing the disclosure of data to unauthorized parties.

Integrity

Integrity refers to protecting information from being modified by unauthorized parties.

Availability

Availability is making sure that authorized parties are able to access the information when needed

Understanding Information Security

- The protection of information and information systems against unauthorized access or modification of information, whether in storage, processing, or transit, and against denial of service to authorized users.
- Information security includes those measures necessary to detect, document, and counter threats to digital and non-digital information.
- Information security processes and policies typically involve physical and digital security measures to protect data from unauthorized access.

Understanding Information Security

- **Goals of Information Security**
  The aim of information security is to develop and maintain a security plan to protect the information.
- **Three main goals of Information security are:**

•**Detection:** The most important element of this strategy is timely detection and notification of a compromise. Intrusion detection systems (IDS) are utilized for this purpose.

•**Prevention:** Security measures must be taken to protect information from unauthorized modification, destruction, or disclosure whether accidental or intentional.

•**Response:** Making important decisions or developing policy while under attack is a recipe for disaster

Why is Cyber Security Important?

Cybersecurity is important because it encompasses everything that pertains to protecting our:

- Sensitive data: Ex. racial or ethnic origin, political opinion, religious beliefs, sexual life or sexual orientation, health data, income tax / property tax data
- Personally identifiable information (PII): name, last name, phone number, vehicle number, driver's license number, photograph, address, voice sample, DOB – POB, IP address, university roll number, edu. qualif.
- Protected health information (PHI):physical and mental health, hospitals, doctors, clinics, psychologists, dentists, nursing homes and pharmacies, ambulance routes and driver details
- Personal information: office location, ATM card, Credit Card, online chats
- Intellectual property: photograph, personal video, thesis, research paper, algorithms.
- Data, Governmental and Industry Information Systems

from theft and damage attempted by criminals and adversaries

- **Don't share personal information on social media (Demo)**

https://x.com/SecurityTrybe/status/1825848612431028380?t=tAXLGVHXPI1HUKqPxhsOUw&s=08

Hacking

- Hacking is the process of identifying and exploiting weakness in a system or a network to gain unauthorized access to data and system resources motivated by financial gain, protest, information gathering (spying), and even just for the "fun" of the challenge.

- It can also be defined as an unauthorized intrusion into the information systems/networks by an attacker by compromising the security.

Example : Exploiting the weakness of default password to gain access to the data stored inside the system.

- Hacking is typically technical in nature but hackers can also use psychology to trick the user into clicking on a malicious attachment or providing personal data

Hacker

- Hacker is very controversial term among programmers arguing that it refers simply to someone with advanced understanding of computers and computer networks. ***But it can range from computer security experts to computer criminal who breaks into Computers***.

Types of Hacker

- **White hat :** A hacker breaks security for non-malicious reasons, either to test their own security system, perform testing , or vulnerability assessment for a client - or while working for a security company which makes security software. The term is generally synonymous with **Ethical Hacker**.

- **Black hat :** Black hat hackers break into secure networks and systems with the motive of destroying, modifying, or stealing some sensitive data, or to make the networks unusable for authorized network users.

- **Grey hat :** A hacker lies between a black hat and a white hat hacker. A grey hat hacker may surf the Internet and hack into a computer system for the sole purpose of notifying the administrator that their system has a security defect. Grey hat hackers sometimes find the defect of a system and publish the facts to the world instead of a group of people.

Ethical Hacking

- Ethical Hacking sometimes called as Penetration Testing is an act of intruding/penetrating into system or networks to find out threats, vulnerabilities in those systems which a malicious attacker may find and exploit causing loss of data, financial loss or other major damages.

- The purpose of ethical hacking is to improve the security of the network or systems by fixing the vulnerabilities found during testing.

- Ethical hackers may use the same methods and tools used by the malicious hackers but with the permission of the authorized person for the purpose of improving the security and defending the systems from attacks by malicious users.

- Ethical hackers are expected to report all the vulnerabilities and weakness found during the process to the management.

Ethical Hacker

- Ethical Hacker is a skilled professional who has excellent technical knowledge and skills and knows how to identify and exploit vulnerabilities in target systems.

- He works with the permission of the owners of systems.

- An ethical Hacker must comply with the rules of the target organization or owner and the law of the land and their aim is to assess the security posture of a target organization/system.

Purpose of Ethical hacking

- The purpose of ethical hacking is to enhance the security of the network or systems by fixing the vulnerabilities found during testing.
- Ethical hackers also utilize the same methods and tools that are used by malicious hackers.
- Ethical hackers take the permission of an authorized person for the purpose of upgrading or improving the security or defending the systems from attacks by malicious hackers or users.
- Ethical hackers are accountable to report all the vulnerabilities and weaknesses found during the process to the management or higher authority.
- Many companies, govt. institutes, CBI, etc. often hire computer experts to hack into their system to find malicious files, vulnerabilities, and weak endpoints so that they can be fixed on time.

Types of Cyber Crimes

- **Cybercrime**, or computer-oriented crime, is a crime that involves a computer and a network.
    - Seeking commercial gain from hacking banks & financial institutions as well a phishing scams & computer ransom ware
- **Cyber Espionage** is the act or practice of obtaining secrets and information without the permission and knowledge of the holder .
    - Using stealthy IT Malware to penetrate both corporate & military data servers in order to obtain plans & intelligence
- **Cyber Warfare** is computer- or network-based conflict involving politically motivated attacks by a nation-state on another nation-state.

Types of Cyber Crimes

- **Cyber terrorism** is the use of the Internet to conduct violent acts that result in, or threaten, loss of life or significant bodily harm, in order to achieve political or ideological gains through threat or intimidation.

- **Cyber Hackivists:** Groups such as "Anonymous" with Political Agendas that hack sites & servers to virally communicate the "message" for specific campaigns.

Targets and Motives

- Corporate
- Defacement
- Takeover/control
- Financial (directly)
- Extortion
- Revenge
- Corporate or personnel image and reputation
- Individual/Personal – Yours and Family
- Governmental/Military
- Political, Religious.

Some Important terms in Hacking

- **Threat:** Anything that has potential to cause harm. There are various threats available to system threats, Network threats, application threats, cloud threats, malicious files threats etc.

- **Vulnerability:** A weakness or a flaw in the system which an attacker may find and exploit. An updated OS, Default Passwords, Unencrypted protocols are all good examples of vulnerabilities.

- **Attack:** Method followed by a hacker/Individual to break into the system. Denial of service attack, mis-configuration attacks, Operating system attacks, Virus, and Worms are all example of Attacks.

- **Attack vectors:** Path or means by an attacker gains access to an information system to perform malicious activities.

Cyber Crime

- Anything done to a person or group of persons which is harmful and disastrous by using the internet or a Computer as a tool / weapon is called Cyber Crime.
- Cyber crimes can involve criminal activities that are traditional in nature, such as theft, fraud, forgery, defamation and mischief, all of which are subject to the Indian Penal Code.
- The following are the list of cyber crime

Cyber Crime

- **According to the United Nation:**
  "Cyber Crime spans not only state but national boundaries as well."

- **In narrows sense:** "Any illegal behavior directed by means of electronic operations that targets the security of computer system and the data processed by them."

- **In a broader sense:** Any illegal behavior committed by means of, or in relation to a computer system or network, including such crimes as illegal possession and offering or distributing information by means of computer system or network.

Categories of Cyber Crimes

- **The Computer as a Target :** In this category using a computer to attack other computers. e.g.
  - Hacking,
  - Virus/Worm attacks,
  - DOS attack etc.
- **The computer as a weapon :** In this category using a computer to commit real world crimes. e.g.
  - Cyber Terrorism,
  - IPR violations,
  - Credit Card frauds,
  - EFT frauds,
  - Pornography, Child Pornography etc.

Distinction b/w Cyber Crime and Conventional Crime

- Traditional criminals usually leave traces of a crime, as fingerprints or other physical evidences. On the other hand, cybercriminals rely on the Internet via which they commit their crimes, and it leaves very little evidence about the crime.

- Since cybercrime involves perpetrators using falsified names and working from remote locations, it usually takes longer to identify the real cybercriminals. In most cases, cybercriminals (such as hackers) escape from arrest because the investigators cannot locate them.

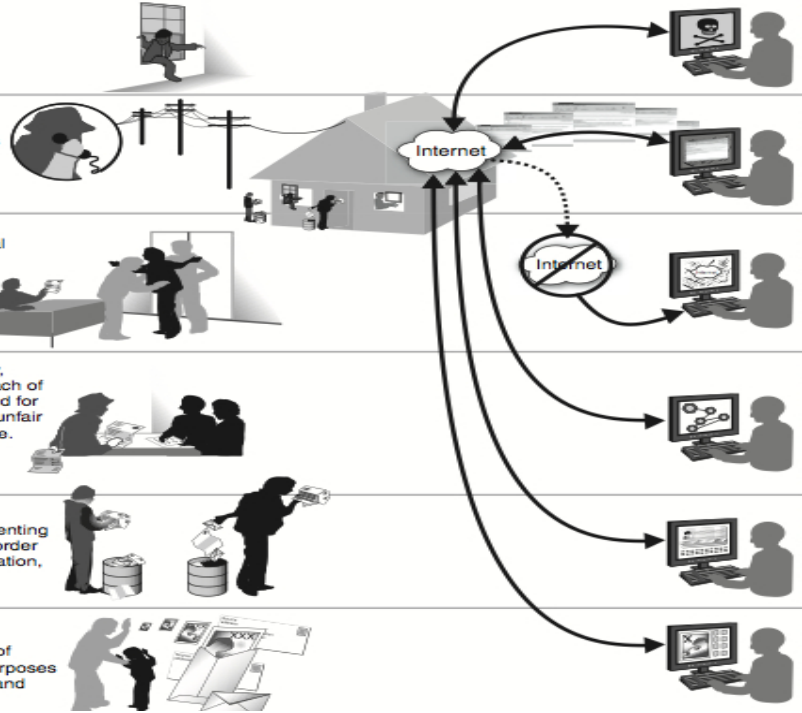
- Traditional crimes take shorter time period to investigate because the criminals usually leave evidence that can be used to spot them. For example traditional criminals can leave evidence such as DNA, fingerprints, photographs and videos captured on or personal belongings such as identity cards, and this makes it easy for investigators to identify and capture the criminals.

- Most of the traditional crimes (such as rape, murder, arson, and burglary among others) involve the use of excessive force that results in physical injury and trauma on the victims. On the other hand, cybercrimes do not require the use of any force since the criminals merely use the identities of their victims to steal from them.

Distinction b/w Cyber Crime and Conventional Crime

Comparison between Traditional Criminal Techniques and Cybercrime

Traditional criminal techniques	Cybercrime
Burglary: Breaking into a building with the intent to steal.	**Hacking:** Computer or network intrusion providing unauthorized access.
Deceptive callers: Criminals who telephone their victims and ask for their financial and/or personal identity information.	**Phishing:** A high-tech scam that frequently uses unsolicited messages to deceive people into disclosing their financial and/or personal identity information.
Extortion: Illegal use of force or one's official position or powers to obtain property, funds, or patronage.	**Internet extortion:** Hacking into and controlling various industry databases (or the threat of), promising to release control back to the company if funds are received or some other demand satisfied.
Fraud: Deceit, trickery, sharp practice, or breach of confidence, perpetrated for profit or to gain some unfair or dishonest advantage.	**Internet fraud:** A broad category of fraud schemes that use one or more components of the Internet to defraud prospective victims, conduct fraudulent transactions, or transmit fraudulent transactions to financial institutions or other parties.
Identity theft: Impersonating or presenting oneself as another in order to gain access, information, or reward.	**Identity theft:** The wrongful obtaining and using of another person's identifying information in some way that involves fraud or deception, typically for economic gain.
Child exploitation: Criminal victimization of minors for indecent purposes such as pornography and sexual abuse.	**Child exploitation:** Using computers and networks to facilitate the criminal victimization of minors.



Type of Network Attacks

Active Network Attack

Attackers not only gain unauthorized access but also modify data, either deleting, encrypting or otherwise harming it.

Passive Network Attack

Attackers gain access to a network and can monitor or steal sensitive information, but without making any change to the data, leaving it intact.

Security threat and Vulnerabilities

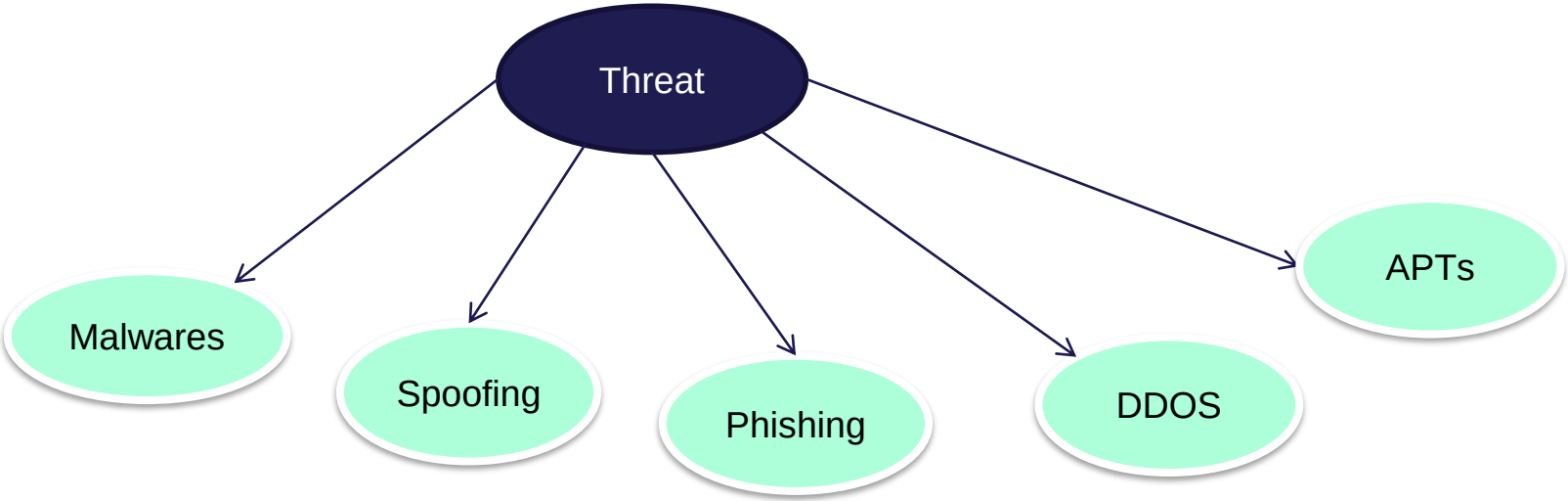
- Threat and vulnerability are terms that are essential to cybersecurity.But oftentimes, we get their meanings confused.

- It is **crucial** for us to understand both threats and vulnerabilities so they can effectively manage the impact of a data compromise and manage cyber risk.

Threat is what an organization is defending itself against, e.g. a DoS attack.

Vulnerabilities are the gaps or weaknesses within software, e.g. an API flaw that lets hackers into a network

Security Threats

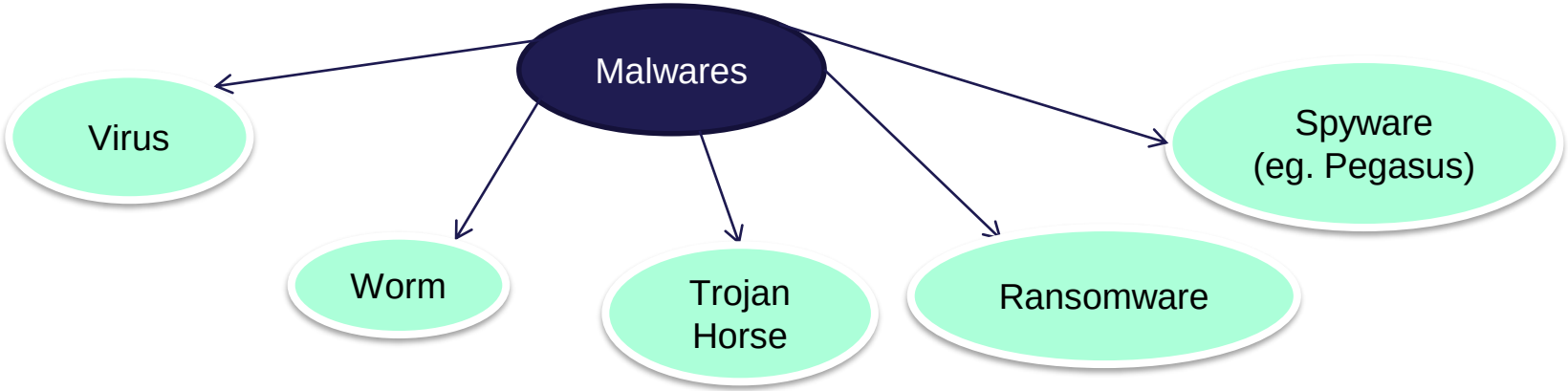


1.Malware

Malware is a file or code typically delivered over a network, that infects, explores, steals or conduct virtually any behavior an attacker wants.

Malware usually has one of the following objectives:

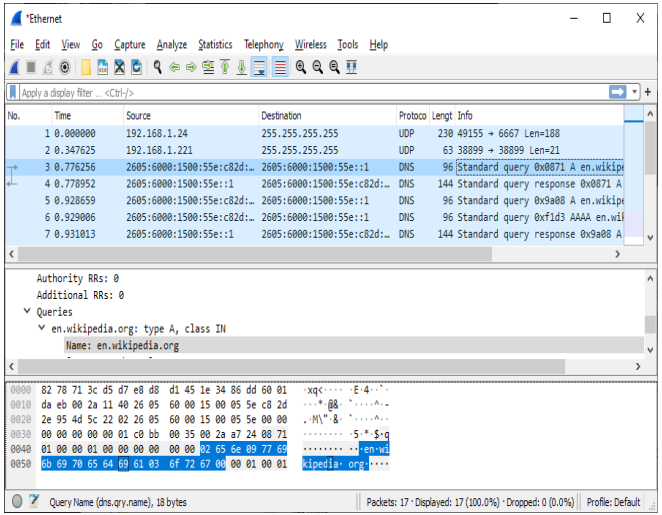
- Provide remote control for an attacker to use an infected machine.
- Send spam from the infected machine to unsuspecting targets.
- Steal sensitive data.



Packet Sniffing

- The act of capturing data packet across the computer network is called **packet sniffing**.
- It is similar to as wire tapping to a telephone network.
- It is mostly used by *crackers and hackers* to collect information illegally about network.

- **ISPs** use packet sniffing to track all your activities such as:
  - What you download.
  - Sites you visit.
  - what you looked on that website.
  - Downloads from a site.
  - Streaming events like video, audio, etc.



2. Spoofing

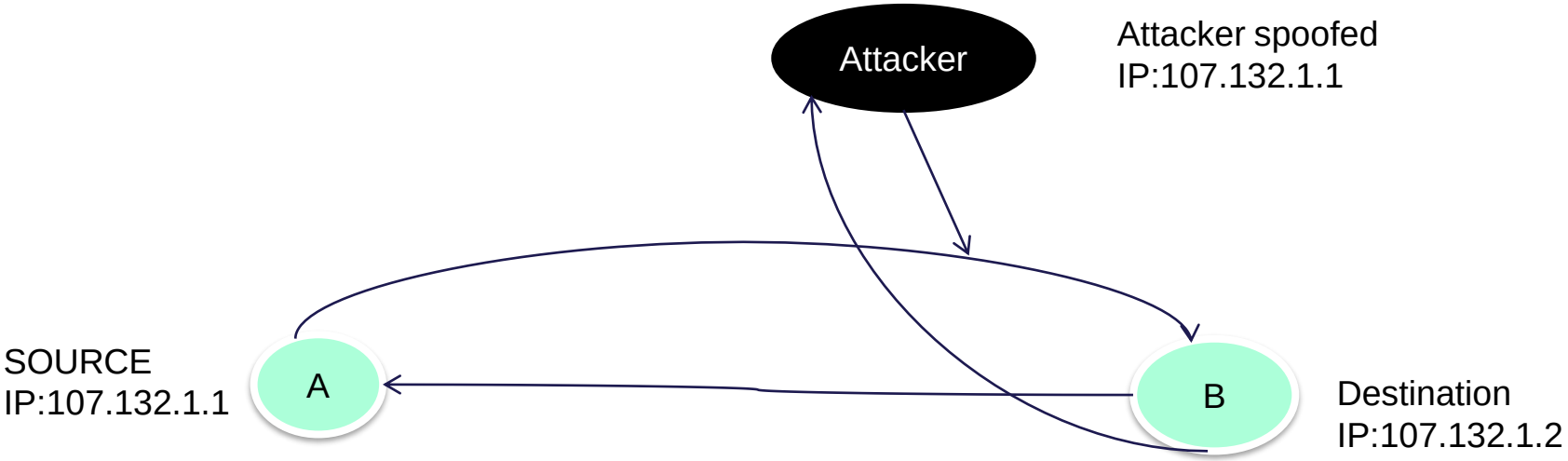
Spoofing : It is a kind of attack where a person steals the details of a legitimate user and acts as another user. It is a kind of identity theft.

Example : Mumbai SIM card fraud

Types of spoofing

- IP spoofing: Attacker uses IP address of another computer to acquire information or gain access from another network.
- Email spoofing: Attacker sends email but makes it appear to come from someone else(reliable email).
- Web spoofing: Attacker tricks web browser into communicating with a different web server than the user intended.

IP Spoofing


Attacker
Attacker spoofed
IP:107.132.1.1
SOURCE
IP:107.132.1.1
A
B
Destination
IP:107.132.1.2

3.Phishing

- **Phishing** is a kind of social engineering attack where a person steals the sensitive information of user in a fraud manner by disguising as a legitimate person.

- By using phishing emails, texts, or social media posts that lead to phishing sites, fraudsters attempt to deceive you into revealing your personal and sensitive information – bank account numbers, credit card info, Social Security number, or login IDs, usernames, and passwords.

Example : Wipro gift card phishing Attack

4.DDOS

- **A Distributed Denial of Service (DDoS)** attack is an attempt to make an online service unavailable by bombarding it with traffic from multiple sources.

- If single source is used then it is called as Denial of Service.

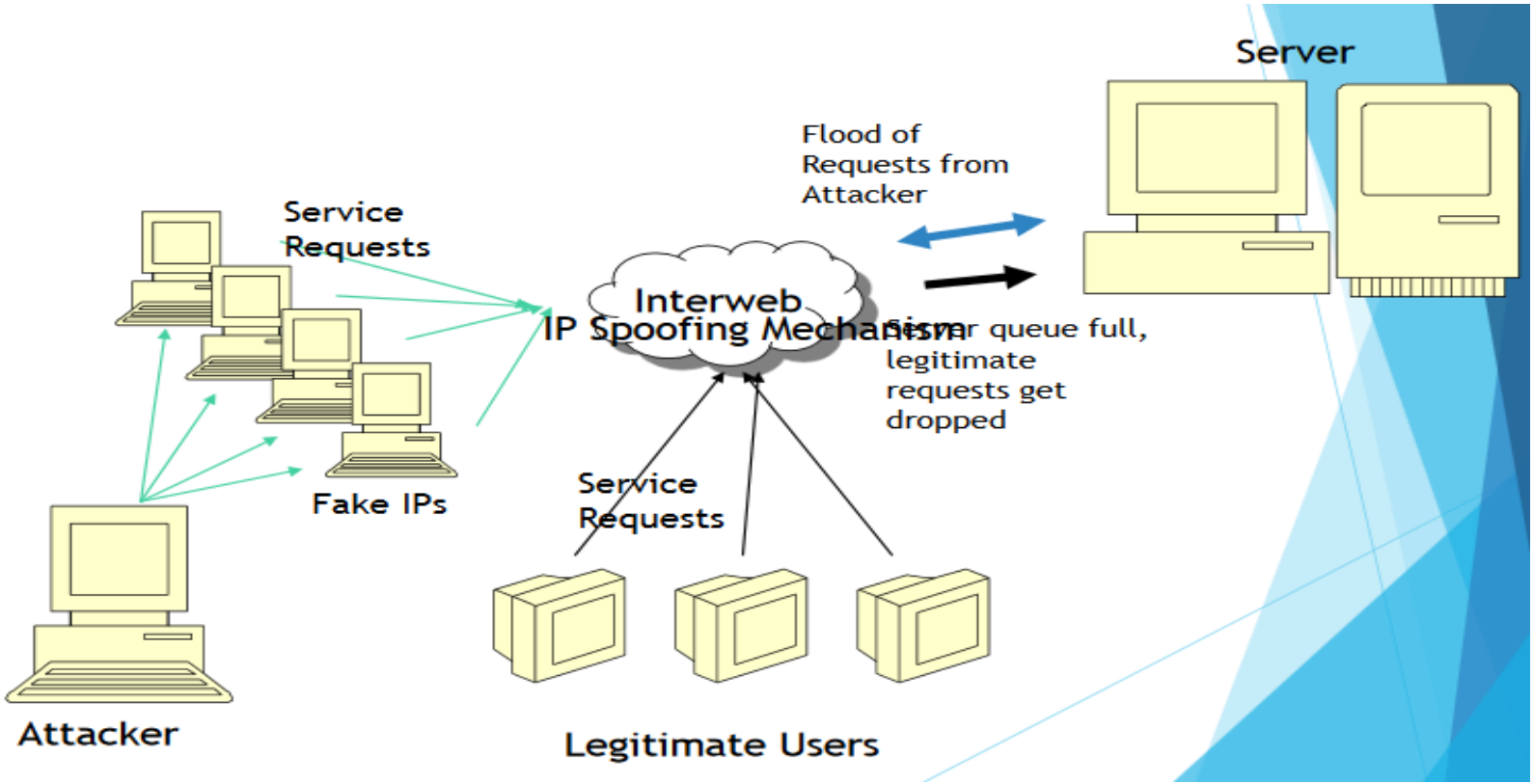
How DDoS attack works ?

- In a typical DDoS attack, the attacker begins by exploiting a computer system and making it the DDoS master.

- The attack master system identifies other vulnerable systems and gains control over them by either infecting the systems with malware or through bypassing the authentication controls.

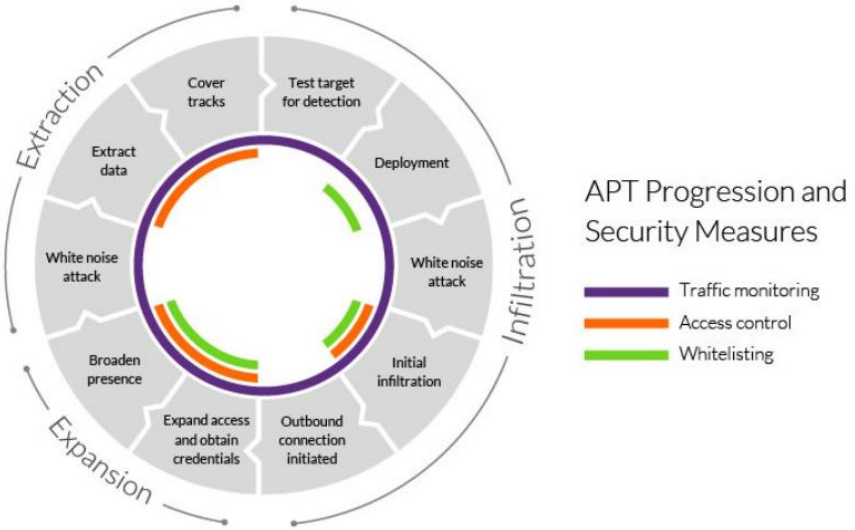
- Then these **botnet** attack victim server.

4.DDOS contd…


Server
Flood of
Requests from
Attacker
Service
Requests
Interweb
IP Spoofing Mechanism
queue full,
legitimate
requests get
dropped
Fake IPs
Service
Requests
Attacker
Legitimate Users

5. APT: Advanced Persistent Threat

- APTs are covert attacks, specifically designed by certain well-established actors with the intention to bypass intrusion detection systems and anti-malware programs.
- So there is a reason that they're called "advanced" attacks: both the malware they use and the character of the danger they present are advanced.
- In order to maintain access for a longer period of time, the hacker tends to use some advanced methods, rewriting malicious script and other sophisticated techniques.



APT Examples: GhostNet

- GhostNet is considered to be one of the most sophisticated and oldest APTs the digital has seen so far. It was first discovered in March of 2009. Its control infrastructure was reported to have been located largely in China, and this attack was directed against the Tibetan community, however, the Chinese government has denied the fact.
- The GhostNet attacks were executed by spear-phishing emails containing malicious downloadable files that loaded a Trojan horse on the user's system, allowing the execution of commands from a remote command and control system, which downloaded malware to take full control of the infiltrated system.
- The malware had the ability to use audio as well as video recording devices to monitor the locations housing the compromised computers. GhostNet was reported to have compromised the devices of political, economic, and media targets in nearly 103 countries, including the embassies of India, South Korea, Indonesia, Romania, and others. The Asian Development Bank and ministries of foreign affairs of Bangladesh, Brunei, Indonesia, Iran, Latvia, Philippines were also among the victims.

APT Examples: Helix Kitten

- Helix Kitten is believed to be an Iran-based adversary group, and this group has been operational since 2014. Its major targets included organizations in aerospace, energy, financial, government, hospitality, and telecommunications, mostly in the Middle East. This advanced group has utilized perfectly structured spear-phishing messages that were so relevant to targeted users.
- As regards the technicalities, this group was most commonly associated with a custom PowerShell implant known as Helminth. The Helminth implant is routinely delivered via macro-enabled Microsoft Excel files demanding user interaction to implement an obfuscated Visual Basic Script. In fact, this is a highly multi-faceted approach: the group made many modifications, downloaded new malware, then manipulated the memory.

Defense and Mitigation

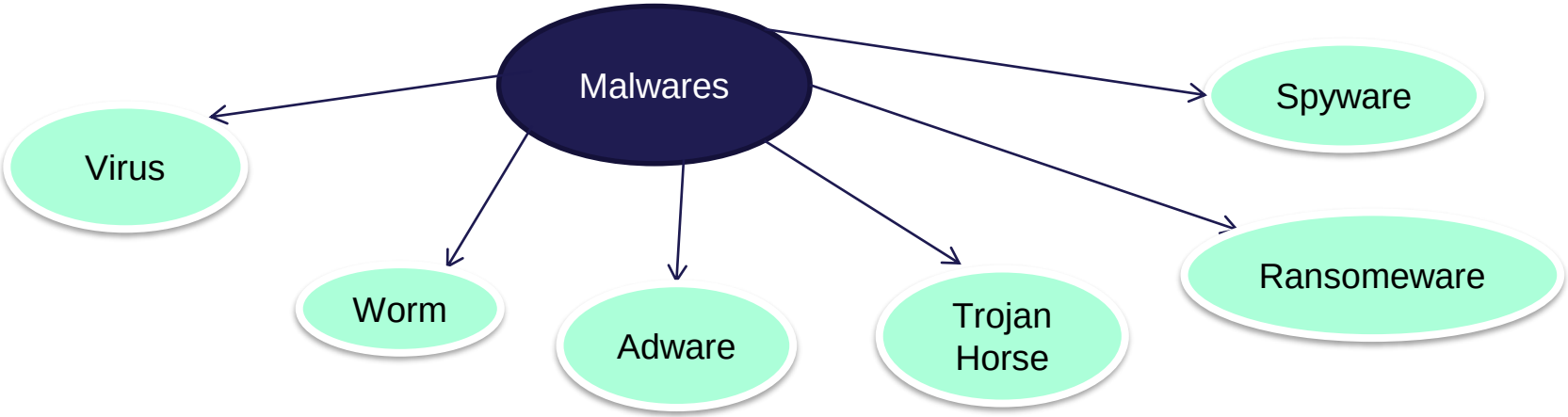
Threat is what an organization is defending itself against, e.g. a DoS attack.

Vulnerabilities are the gaps or weaknesses within software, e.g. an API flaw that lets hackers into a network.

- **Security Defense** is an approach to **cyber security** in which a series of **defensive** mechanisms are layered in order to protect valuable data and information.

- **Vulnerability Mitigation** refers to policies and processes put in place by companies to help prevent security incidents and data breaches as well as limit the extent of damage when security attacks do happen.

1. Malwares Defense



1. Malwares Defense - Antivirus

- **Virus Signature:** Security experts study the infected files looking for code fragments that are unique to the particular computer virus.
  Example: A string of sequence of instruction specific for each virus.

- Once they have located such a set of characteristic instructions, they can construct a character string that uniquely identifies this virus. This character string is known as a signature for the virus.

- All the signatures together create the malware database.

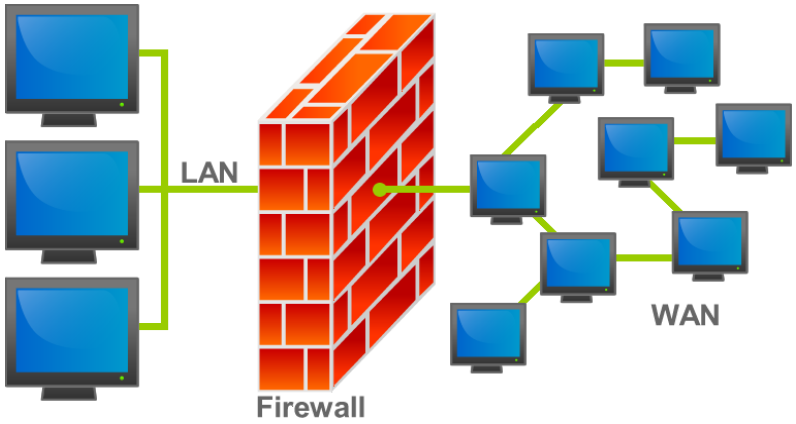
- Its a kind of digital fingerprint.

- AntiVirus scan compare the analyzed object with a database of signatures.

- AntiVirus software packages have to be frequently updated, so that they always are using the most up- to- date database of virus signatures.

2. Spoofing Defense

1) Packet-filtering device : Packet filtering is a firewall technique used to control network access by monitoring outgoing and incoming packets allowing them to pass or halt based on the source and destination Internet Protocol (IP) addresses, protocols and ports.
Example : Network layer Firewall/proxy server .



2) IPv6 is the newest Internet Protocol version. It makes IP spoofing harder by including encryption and authentication steps.

3.Phishing Defense

- Educate your employees and conduct training sessions with mock phishing scenarios.
- Deploy a SPAM filter that detects viruses, blank senders, etc.
- Keep all systems current with the latest security patches and updates.
- Install an antivirus solution, schedule signature updates, and monitor the antivirus status on all equipment.
- Deploy a web filter to block malicious websites.
- Encrypt all sensitive company information.
- Require encryption for employees that are telecommuting.

4.DDOS Defense - IDIP

IDIP: Intruder Detection and Isolation Protocol –

Following three process are involved in this :

1)Track : When an attack traverses an IDIP-protected network, each IDIP node along the path is responsible for auditing the connection;

2) Distributed : When a component detects an intrusion attempt, the detector distributes an attack report to its neighbors who can then help trace the attack path and respond to the intrusion. these neighbors further distribute the attack description along the path of the attack.

3)Block : Request are blocked by using firewall.

5. APT: How to be protected?

- **Use Strong Passwords**: Create strong, unique passwords for each account and change them often.
- **Enable Two-Factor Authentication**: Add extra security by using two-factor authentication (2FA), which requires a second verification step, like a text code.
- **Keep Software Updated**: Regularly update your operating system, apps, and security software to fix weaknesses.
- **Install Antivirus**: Use good antivirus software to detect and block threats.
- **Use Firewalls**: Set up firewalls to block unauthorized access to your network.
- **Back Up Data**: Regularly back up important files to an external drive or cloud storage to prevent data loss.
- **Monitor Activity**: Keep an eye on your network for any unusual activity.
- **Limit Access**: Only give access to sensitive information to those who need it and restrict administrative rights.
- **Secure Remote Access**: Use secure methods like VPNs when accessing your network remotely.

Latest Tools Used in Cyber Crimes

- **Dark Web Frameworks**: Criminals use frameworks like Tor, I2P, and Freenet to access the dark web, facilitating anonymous communication and transactions.
- **Malware-as-a-Service (MaaS):** Pre-built malware kits, such as Emotet and TrickBot, are sold on dark web marketplaces, making it easier for attackers to launch campaigns.
- **Ransomware**: Tools like REvil, GandCrab, and Maze ransomware are used for extortion.
- **Social Engineering Kits**: Pre-built phishing kits, like Phishing Frenzy and KingPhisher, simplify social engineering attacks.
- **Crypters and Packers**: Tools like UPX and VMProtect are used to obfuscate malware.

Latest Techniques Used in Cyber Crimes

- Spear Phishing: Targeted phishing attacks using personalized emails or messages.
- Business Email Compromise (BEC): Scammers impersonate executives or employees to trick victims into transferring funds.
- Advanced Persistent Threats (APTs): Sophisticated, multi-stage attacks targeting organizations.
- Zero-Day Exploits: Attackers exploit previously unknown vulnerabilities.
- Deepfakes: AI-generated fake audio, video, or images used for social engineering.
- Fileless Malware: Malware resides in memory only, evading traditional detection.
- Living-off-the-Land (LotL): Attackers use legitimate system tools to carry out malicious activities.
- Domain Name System (DNS) Tunneling: Encrypted communication through DNS.
- Cloud jacking: Compromising cloud services to steal data or launch attacks.
- Artificial Intelligence (AI) and Machine Learning (ML) attacks: Using AI/ML to evade detection, automate attacks, or create sophisticated phishing campaigns.

Thankyou ☺